Women Safety Route App: A Data-Driven App for Safe Route Navigation in Delhi

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***Abstract*—Urban cities like Delhi with exceptionally high population and extremely high crime rates need solutions towards safer travel especially for women considering the high number of rapes and non-domestic crimes happening against women [?]. Delhi reached a crime rate of 145 cases per 100,000 women, which is significantly above the national average of 67 and much higher than any Indian state. My project finds the safest route by inputting the source location and the destination location by using Python's geospatial libraries and data science libraries, overlaying a 500m $\times$ 500m grid all over Delhi, with each grid assigned a safety score using a crime dataset that included crimes committed in Delhi and the Delhi boundary geodataframe. The safest route was calculated using the Dijkstra algorithm. The project also shows police stations as pop-ups which is really helpful in case of emergencies and mindful for a safety app. This app generously contributes to safer urban navigation, aiming to contribute towards safer mobility, ensuring women can make informed and secure travel decisions.**

Keywords: Women Safety, Safe Routing, Dijkstra Algorithm, Geospatial Analysis, Crime Mapping

I. INTRODUCTION

Given the rising incidence of crime in Delhi, it is imperative to prioritize the safety of its citizens. The city consistently experiences a high crime rate across various categories. Traditional routing systems often overlook safety considerations, focusing solely on the shortest path. However, identifying the safest route is of greater significance than merely determining the shortest one. Delhi consistently has the highest crime rate per capita and per person (actually 5 per person) and frequently tops the charts for certain categories like kidnapping and robbery. This shows how unsafe the city environment can be for its citizens.

The records also support this. Delhi has reported 118,822 total complaints in 2025, 129,693 in 2024, and 160,931 in 2023. Non-domestic crimes against women are particularly concerning, with 1,393 cases registered in the first quarter of 2023, 1,393 in the same period of 2024, and 1,199 in 2025. Because of such conditions, it becomes a necessity for a city like Delhi to have a platform that can suggest the safest route instead of just the shortest one. Women often feel unsafe while travelling in the evening; it becomes even more concerning at night. Morning travel also does not guarantee safety, so while the government is making efforts in addressing these issues, we can make technological solutions such as applications that suggest the safest possible routes. Also, safety is not only a matter of convenience but a fundamental right that every citizen wants and should definitely be provided, especially in metropolitan cities like Delhi where women's safety is a major concern.

As we look at the already existing mobile apps and tech solutions like the **Himmat app** developed by Delhi Police especially for women, which enables women to send emergency SOS with location, **112 India** also allows users to send an emergency SOS alert with integration to official emergency services, and **Safetipin** which provides safety-related information about Delhi neighbourhood, none of these directly solves the problem of safer travel. Most of them are useful after you get into trouble, but why not be precautious and smart and choose the safest route? My project specifically targets women's safety by suggesting the safest route from one location to another. I have done this by analyzing previous years' crime reports and highlighting nearby police stations as pop-ups. Unlike traditional algorithms that only show crime density heatmaps without route optimization, my algorithm uses the **Dijkstra method** to find the safest route while also showing police stations, so the user knows where to go if something goes wrong. A unique safety score is calculated for each grid in the city, making the route optimization data-driven and reliable. The app also visualizes crime-prone zones on interactive maps and generates comparative charts for better understanding of crime distribution.

II. LITERATURE REVIEW

II-A. Previous Works

1) First research paper

Aliasgar Eranpurwalal, Fatema Indorewala2, Nafisa Mapari3, Saloni Mishra proposed a research paper regarding the safety of women and safe route prediction which includes features like distress calls, emergency buttons, and sharing location [?]. The location and routing is done using the Google Maps API. The key features of the paper are distress calls, basic route suggestions, location sharing with family members, estimation of arrival time, and alert messages to family. But the routing is done on the Google Maps traffic or distance, not on the crime data, which makes it a less preferable option for navigating a safer route. My work addresses this gap by using actual crime data, and the safest route is computed using graph travel algorithms. The project more over focuses on emergency after danger has happened but my project focuses on preferring the safest routes.

2) Second research paper

KUSHAL AGRAWAL, AVIRAL SRIVASTAVA, et al. proposed a research paper in 2024 named **SafeRoutes** which potentially targeted finding safer routes for women [?]. They integrated crime data, unsupervised learning, and real-time cab/mapping services. The project integrated advanced use of AI and unsupervised learning. It's data-driven and has a real-time approach. The methodology used is also strong with data ingestion pipelines, clustering, and integration with map APIs. But the project is more focused on design and framework rather than practical implementation. The project is broad and theoretical instead of being implementable in real life. My project is more useful in real life as I am integrating Delhi crime dataset with routing. My project is more user-facing rather than just a framework.

3) Third Research paper

Dr Jogendra Kumar Nayak and Danish Benazeer (IIT Roorkee) proposed a research paper whose objective was to analyse the issues that affect women safety in Delhi's public transport and suggest solutions [?]. The approach of the project was to conduct surveys, questionnaires, and assessments. And it's more focused on the fear of victimization and categorized women on travel patterns. This project proposes the authorities to identify risk spot zones based on this data and improve safety measures in transportation systems. This paper identifies problems and suggests improvements for the authorities and depends on surveys that get outdated, but women do not directly benefit from it while travelling, but my app helps them to directly make safer travel decisions.

II-B. Existing Safety Applications

- **112 India**: This is a government national emergency app that allows quick SOS alerts, connection with police, and tracking location [?]. It is available in 23 states and integrates with ERSS systems to provide unified rapid responses.
- **Bsafe**: This app allows live location sharing and automatic audio and video sharing in case of emergencies and also facilitates with voice-based SOS alert systems. It works globally and is widely adopted.
- **Safetipin**: This app maps and scores the public spaces on safety and suggests the safe travel routes and also helps users rate and share information about their localities.
- **Himmat app**: This is a Delhi Police app that facilitates instant SOS alerts and also audio and video streaming to police with location sharing for faster tracking in case of emergencies [?].
- **Shake2Safety**: Sends emergency alerts by shaking the phone or pressing the power button, works offline and notifies registered contacts.

II-C. Research Gaps

- The existing apps focus more on the emergency response after an incident has happened and not on the idea of finding the safe routes.
- Many are focused worldwide and or are more generic and focus on India or the world, but my app is specifically for Delhi which makes it more detail-oriented like using a crime dataset of Delhi or Delhi Metro dataset.
- Hardly any system uses real-time crime data and routing across Delhi together.
- Most of the prior projects depend on Google Maps or other types of routing that focus on traffic time and distance which typically targets shorter and more comfortable routes, but my project's core focus is to suggest the safest route.
- Many research papers propose AI/ML frameworks (clustering, unsupervised learning) but remain at the design/theoretical level. They do not translate into user-friendly applications deployable in real time.

III. METHODOLOGY

III-A. Data Collection

1) **Crime Dataset** (`crime_dataset (1).json`): This JSON dataset was obtained from Kaggle. It contains attributes such as `crime_number`, `crime_type`, `longitude`, `latitude` (of the location of crime), `occurred` (which tells about the date and time of occurrence of the crime), and `disposition` (current status of the case). This dataset helped in the safety score calculation which eventually helped in plotting the Delhi safety heatmap and implementing the Dijkstra algorithm.
2) **Delhi Boundary Dataset** (`Delhi_Boundry.geojson`): This publicly available GeoJSON dataset was obtained to make plotting of the boundary of Delhi. It contains attributes `type`, `name`, `crs`, and `features` (which contains the coordinates of the Delhi boundary). This dataset helped in plotting the boundary of Delhi for further use, that is, creating the heatmap and plotting the safest route.
3) **Delhi Police Stations Dataset** (`delhi_police_station_locs`): This dataset is a KML file which contains the columns `Police Station` (names), `longitude`, and `latitude` (of the respective

police station). This dataset helped to mark police station pop-ups.

4) **Delhi Metro Dataset**: This is an Excel file which contains columns `ID (Station ID)`, `Station Names`, `Dist. From First Station(km)`, `Metro line`, `longitude`, `latitude`, `layout`, and `opened(year)`. This dataset helps to make the dataset broader by merging it with the crime dataset.

III-B. Pandas Libraries Used

1) **Data Handling & Processing**: `pandas`, `numpy`, `json`, `xml.etree.ElementTree` (JSON/XML), `base64`, `io`
2) **Geospatial & Mapping**: `geopandas`, `shapely`, `folium`, `streamlit-folium`
3) **Graph & Routing**: `Networkx`
4) **Visualization**: `matplotlib`
5) **Web Application & Interface**: `streamlit`, `webbrowser`

IV. IMPLEMENTATION APPROACH

IV-A. Data Preprocessing

- The Crime dataset (JSON file) was imported and read. `longitude` and `latitude` were changed to `Longitude` and `Latitude` to match the columns of the Delhi metro dataset. The longitude and latitude values were rounded off up to 2 decimal places, and a dataframe named `df` was created.
- The Delhi metro dataset (CSV file) was imported and read. The `Longitude` and `Latitude` columns were rounded off up to 2 decimal places, and a dataframe named `location_df` was created.
- Then both `df` and `location_df` were merged, and a new dataframe named `merged_df` was created.
- A KML file was imported and converted into a CSV file. The data in it was cleaned by splitting the coordinates by comma (`,`) and appending the values on the left of the comma (index 0) into the latitudes list and on the right of the comma into the longitude list. The values in the stations column were cleaned by stripping unnecessary spaces, and then a dataframe with these three lists named `df_kml` was created.

IV-B. Grid Creation (Delhi Boundary and Safety Grid)

- A GeoJSON file (`delhi_boundry`) was loaded which was converted into a coordinate reference system in meters.
- Min and max latitude and longitude were stored in variables `min_x`, `min_y`, `max_x`, and `max_y` using `total_bounds`.
- A list named `grid_cells` was created which contains shapely polygons (boxes) in the range `min_x`, `min_y`, `max_x`, `max_y`.
- A geodataframe was created named `grid_gdf` with geometry `grid_cells`, and then `grid_gdf` and `gdf_boundry` were overlayed.
- `merged_df` (a dataframe) was converted into a geodataframe named `gdf_crime` so that it could be plotted further.
- `Gdf_crime` and `gdf_boundry` geodataframes were spatially joined with `predicate = "within"`. This joins each crime point with the grid cell inside which the point lies. This is to calculate the crimes conducted in each grid area.

IV-C. Safety Score Calculation

- After the join, all the crime columns with their grid cell ID are present, and now each grid cell has its crime counts, and the size of each grid cell is equal to the number of crimes that happened in it.
- The safety scores were calculated, which was basically normalizing the values of the crime count to get all the values between 0 and 1.
- The formula used is:

$$\text{Safety\_score} = 1 - \frac{\text{crime\_count}}{\text{max\_count}}$$

- Grid cells with no crimes were assigned the value 0, and all the grid cells were assigned the safety scores.
- A graph was initialized in which each grid cell was added as a node with two attributes: its geometry and safety score.
- A spatial index (`sindex`) was created to speed up neighbor searches.
- For each grid cell, the bounding boxes that intersect it were found, and then only the ones whose boundary actually touches the cell boundary were kept.
- The weight of the edges was calculated based on the safety of the adjacent cells:

$$\text{Weight} = 1 - \left(\frac{\text{safety\_score cell 1} + \text{safety\_score cell 2}}{2} \right)$$

- A function `nearest_node` was used to map the user inputs (start and destination) into the grid, which identifies the grid cell whose geometry is closest to the given point. So now the start and end locations are marked on the graph.

IV-D. Path Finding

Finally, **Dijkstra's algorithm** was applied on the constructed graph using `NetworkX`'s `shortest_path` function to calculate the safest path between the start node and the end node.

IV-E. Geospatial Mapping Workflow

- A GeoJSON file was loaded of Delhi police stations with their locations.
- A `folium` map was initialized on Delhi. Each of the grid cells was colored according to its safety score: dark green for the safest grid with 0 crime counts and red for the most unsafe grids with the maximum number of crime counts.
- If a path was found, the grid was reprojected to EPSG:4326 (WGS84) for map compatibility.
- The centroids of the grid cells along the path were extracted as route coordinates.
- If a path was found, a blue line was plotted to represent the safest route.
- The start and end coordinates were given green and red markers, respectively.
- The police station dataset was iterated row by row, and a blue marker with the police station name was added to the map.
- The map was saved as an HTML file (`delhi_police_stations_only.html`) and then opened automatically in a web browser.
- A Folium map was initialized and centered at Delhi (coordinates: 28.6139, 77.2090) with a suitable zoom level.
- The Delhi boundary was added using a GeoJson layer for spatial reference. The boundary was drawn with a black outline, weight = 2 and transparent fill. This was like a base map over which police station markers were later plotted.

IV-F. Visualization Part

1) **Bar Graph (crime type distribution)**: `matplotlib` bar plot was used to plot the different types of crime and their frequency.
2) **Scatter Plot (crime locations)**: `matplotlib` scatter plot was used to plot the locations as dots on the exact location on the plot between longitude and latitude.
3) **Safety Score Map (Grid + route)**: Delhi boundary map was drawn with grid cells color-coded on the safety score. Start and End points were marked with green and red colors, respectively, and then the safest route was shown with a blue line. It was drawn using `matplotlib`, and then the image was encoded to base64, and an HTML pop-up was created and opened in a different browser.
4) **Police Station Map**: Police station coordinates were used to mark the police stations with blue pop-ups, and then this `folium` map was saved in an HTML file and opened in a browser.

V. RESULTS

The paper presents the following visualization results:

1) A Graphical User Interface (GUI) where the user can enter the start and end coordinates to find the safest route.

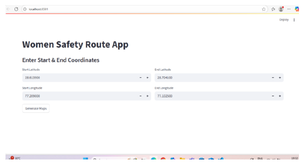

Fig. 1. GUI for safest route input

2) A Bar graph plotting of different types of crimes and their frequency (*Crime Type Distribution in Delhi*).

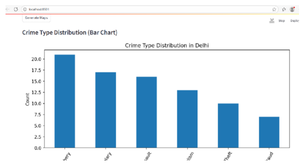

Fig. 2. Crime Type Distribution in Delhi

3) A Scatter plot that consists of crimes marked at their respective longitude and latitude (*Crime Locations*).

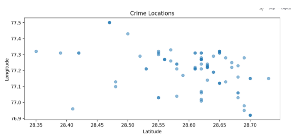

Fig. 3. Crime Locations in Delhi

4) A Delhi map with color coded by safety score and safest route according to the start and end coordinates given earlier.

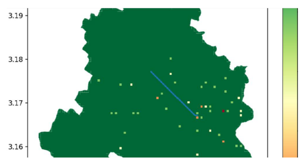

Fig. 4. Delhi Map with Safety Score and Safest Route

5) A map showing Police stations as pop-ups.

VI. CONCLUSION

The goal of the project is to find the safest route for women in order to make their travel safer. This in my project was done by using a crime dataset of Delhi and overlaying a grid of 500m in size all over Delhi and assigning each grid a safety score and then calculating the safest route using the **Dijkstra algorithm**. Police stations are shown as blue coloured pop-ups and also comparative plots were made for different types of crimes happening in Delhi. The project can really contribute to being a safety tool for women while

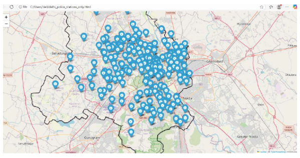

Fig. 5. Delhi Police Stations Displayed as Pop-ups

travelling. In the future, real-time crime data can be integrated into the project, and it can also be expanded to other cities.

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